

Chat Link: <https://parley.mit.edu/share/fxKLxTyAmK7wtGXmwG3Lv>

```
global_data={
  "attribute_1": "total_cost",
  "attribute_2": "material_continuity",
  "attribute_3": "conductivity_MS_per_m",
  "attribute_4": "manufacturing_joining_compatibility",
  "attribute_5": "mass_kg",

  "direction_attribute_1": "negative",
  "direction_attribute_2": "positive",
  "direction_attribute_3": "positive",
  "direction_attribute_4": "positive",
  "direction_attribute_5": "negative",

  "attribute_1_min": 5,
  "attribute_1_max": 100,
  "attribute_2_min": 0,
  "attribute_2_max": 1,
  "attribute_3_min": 20,
  "attribute_3_max": 65,
  "attribute_4_min": 0,
  "attribute_4_max": 1,
  "attribute_5_min": 0.5,
  "attribute_5_max": 15,

  "copper_price_per_ton": 9500,
  "aluminum_price_per_ton": 2500,
  "copper_price_per_kg": 9500,
  "aluminum_price_per_kg": 2500
}
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regional_data=[
  {
    "region": "United States",
    "copper_product_market_share": 0.93,
    "aluminum_product_market_share": 0.07,
    "weight_attribute_1": 3,
    "weight_attribute_2": 4,
    "weight_attribute_3": 3,
    "weight_attribute_4": 5,
    "weight_attribute_5": 1
  },
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{
  "region": "China",
  "copper_product_market_share": 0.65,
  "aluminum_product_market_share": 0.35,
  "weight_attribute_1": 5,
  "weight_attribute_2": 3,
  "weight_attribute_3": 3,
  "weight_attribute_4": 3,
  "weight_attribute_5": 2
},
{
  "region": "India",
  "copper_product_market_share": 0.55,
  "aluminum_product_market_share": 0.45,
  "weight_attribute_1": 5,
  "weight_attribute_2": 3,
  "weight_attribute_3": 2,
  "weight_attribute_4": 2,
  "weight_attribute_5": 2
},
{
  "region": "Europe",
  "copper_product_market_share": 0.90,
  "aluminum_product_market_share": 0.10,
  "weight_attribute_1": 3,
  "weight_attribute_2": 4,
  "weight_attribute_3": 4,
  "weight_attribute_4": 4,
  "weight_attribute_5": 1
}
]

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```

product_data=[
  {
    "region": "United States",
    "dominant material": "cu",
    "product": "Type NM-B Copper Building Wire (12/2, 100 ft)",
    "copper_kg": 2.7,
    "aluminum_kg": 0,
    "total_mass_kg": 3.5,
    "non_material_cost_per_unit": 18,
    "attribute_2_value": 1,
    "attribute_3_value": 58.0,

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"attribute_4_value": 1.0,
"attribute_5_value": 3.5
},
{
  "region": "United States",
  "dominant material": "al",
  "product": "Type NM-B Aluminum Building Wire (10/2, 100 ft)",
  "copper_kg": 0,
  "aluminum_kg": 1.3,
  "total_mass_kg": 2.2,
  "non_material_cost_per_unit": 18,
  "attribute_2_value": 0,
  "attribute_3_value": 35.0,
  "attribute_4_value": 0.5,
  "attribute_5_value": 2.2
},
{
  "region": "China",
  "dominant material": "cu",
  "product": "Type NM-B Copper Building Wire (12/2, 100 ft)",
  "copper_kg": 2.7,
  "aluminum_kg": 0,
  "total_mass_kg": 3.5,
  "non_material_cost_per_unit": 10,
  "attribute_2_value": 1,
  "attribute_3_value": 58.0,
  "attribute_4_value": 1.0,
  "attribute_5_value": 3.5
},
{
  "region": "China",
  "dominant material": "al",
  "product": "Type NM-B Aluminum Building Wire (10/2, 100 ft)",
  "copper_kg": 0,
  "aluminum_kg": 1.3,
  "total_mass_kg": 2.2,
  "non_material_cost_per_unit": 10,
  "attribute_2_value": 0,
  "attribute_3_value": 35.0,
  "attribute_4_value": 0.5,
  "attribute_5_value": 2.2
},
{
  "region": "India",
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"dominant material": "cu",
"product": "Type NM-B Copper Building Wire (12/2, 100 ft)",
"copper_kg": 2.7,
"aluminum_kg": 0,
"total_mass_kg": 3.5,
"non_material_cost_per_unit": 7,
"attribute_2_value": 1,
"attribute_3_value": 58.0,
"attribute_4_value": 1.0,
"attribute_5_value": 3.5
},
{
  "region": "India",
  "dominant material": "al",
  "product": "Type NM-B Aluminum Building Wire (10/2, 100 ft)",
  "copper_kg": 0,
  "aluminum_kg": 1.3,
  "total_mass_kg": 2.2,
  "non_material_cost_per_unit": 7,
  "attribute_2_value": 0,
  "attribute_3_value": 35.0,
  "attribute_4_value": 0.5,
  "attribute_5_value": 2.2
},
{
  "region": "Europe",
  "dominant material": "cu",
  "product": "Type NM-B Copper Building Wire (12/2, 100 ft)",
  "copper_kg": 2.7,
  "aluminum_kg": 0,
  "total_mass_kg": 3.5,
  "non_material_cost_per_unit": 15,
  "attribute_2_value": 1,
  "attribute_3_value": 58.0,
  "attribute_4_value": 1.0,
  "attribute_5_value": 3.5
},
{
  "region": "Europe",
  "dominant material": "al",
  "product": "Type NM-B Aluminum Building Wire (10/2, 100 ft)",
  "copper_kg": 0,
  "aluminum_kg": 1.3,
  "total_mass_kg": 2.2,
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"non_material_cost_per_unit": 15,  
"attribute_2_value": 0,  
"attribute_3_value": 35.0,  
"attribute_4_value": 0.5,  
"attribute_5_value": 2.2  
}  
]
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